ROSE5770 Individual Assessment

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Question 1

English main text: 654 words.

(1) Framework and data transfer

An intelligent autonomous unmanned system turns a human objective into physical action through connected sensing, planning and control loops. A ground control station (GCS), with a human-machine interface (HMI), accepts the user's objective, operating constraints and intervention commands. Mission planning, potentially assisted by artificial intelligence, decomposes this objective into tasks, determines their sequence and completion conditions, and passes the current task to motion planning (Chen, 2022, pp. 21-22).

Executing these tasks requires an updated understanding of the vehicle and its surroundings. Sensors on the hardware platform supply measurements to perception and positioning. Perception extracts obstacles and relevant objects, while state estimation produces pose and velocity. Simultaneous localisation and mapping (SLAM) jointly estimates the vehicle's pose and constructs or updates an environmental map (Cadena et al., 2016). Estimated states feed both planning and control; maps, obstacle information and target locations inform planning.

With this information, motion planning converts the assigned task into a collision-free route and a time-parameterised trajectory. Desired position, velocity, acceleration and yaw become references for flight control, subject to obstacle clearance and vehicle limits (Mellinger & Kumar, 2011). This connection also carries feedback: an infeasible route, changed obstacle or completed movement can prompt mission planning to revise the task sequence. Thus, the task planner decides what to do, while the motion planner determines how movement can achieve it.

The controller then converts these references into actuator commands using state feedback and a model relating inputs to vehicle dynamics. For an unmanned aerial vehicle (UAV), the outer loop tracks position and velocity, while the inner loop stabilises attitude (Chen, 2022, p. 20). Motors and other actuators move the hardware platform, whose resulting motion produces new sensor measurements. This closes the physical feedback loop and allows the controller to correct tracking errors and respond to disturbances.

Alongside vehicle control, application data processing converts observations into useful outputs, such as inspection images or delivery records. Communications carry commands from the GCS and return results, progress and warnings to the user (Chen, 2022, p. 21, Fig. 2). Remaining energy and communication availability must also inform mission decisions, connecting the supporting hardware and communication functions to the feasibility of continued operation.

(2) Levels of autonomy

Within this framework, autonomy depends on how decision responsibility is shared between people and the system. The four levels in the U.S. Department of Defense taxonomy describe this allocation (U.S. Department of Defense, 2011, p. 46, Table 3). Applying them to familiar operating examples, Level 1, Human-Operated, places all operational decisions with the human. A remotely piloted model

aircraft illustrates this when its pilot continually selects the route, manoeuvres and landing actions. The communication link lets the pilot control the aircraft from a distance.

At Level 2, Human-Delegated, the operator authorises particular automated functions. A camera drone pilot may activate automatic take-off or position holding while deciding the overall flight sequence and when each function should stop. Responsibility has therefore shifted for an individual operation, but the human still coordinates the mission and selects the next action.

Level 3, Human-Supervised, extends this delegation to sensor-informed execution of directed tasks under human oversight. For example, a surveying UAV can carry out an approved imaging mission while its operator monitors progress and provides guidance. The system may adjust its actions within the permitted task, and the operator can initiate changes or intervene. Supervision therefore remains part of normal mission operation.

At Level 4, Fully Autonomous, the human specifies goals and the system translates them into tasks and executes them without routine human interaction. A mapping robot vacuum illustrates this within a bounded indoor cleaning task when it plans coverage, avoids obstacles and returns to its charger. Emergency intervention or changes to the goals remain possible under the definition (U.S. Department of Defense, 2011, p. 46). Consequently, autonomy should be assessed for a specified task and operating domain: the same vehicle can use different levels in different modes, and automatic control alone does not establish full autonomy.

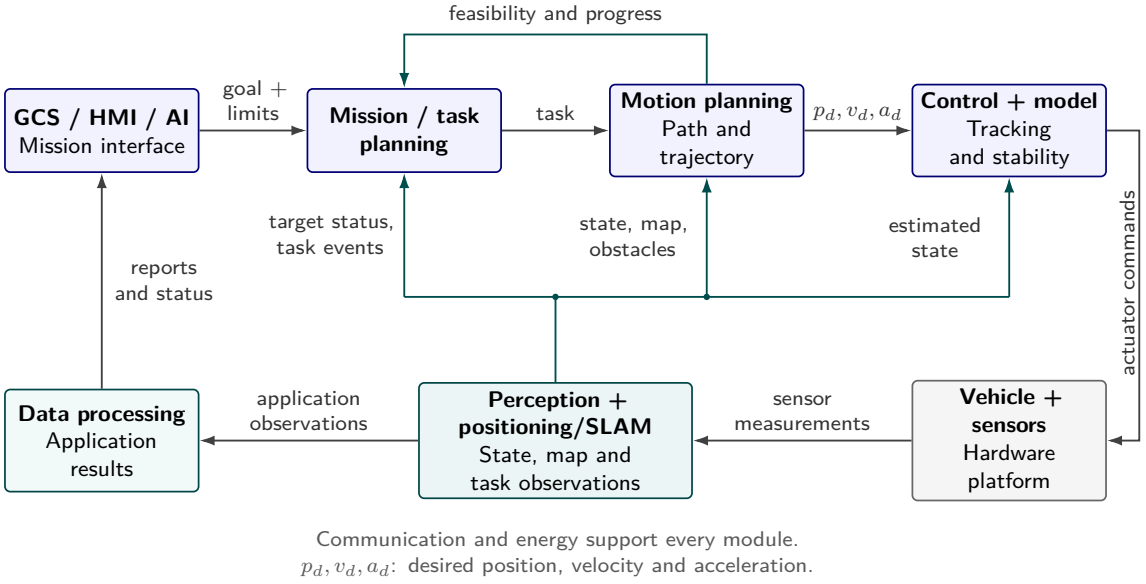


Figure 1: Original schematic based on the autonomous-system architecture in Chen (2022, p. 21, Fig. 2). Arrows show information types and decision feedback.

Table 1: Four autonomy levels from U.S. Department of Defense (2011, p. 46, Table 3). The operating examples are illustrative applications of the taxonomy.

Level	Decision responsibility	Illustrative example
1. Human-Operated	Human chooses actions and directly controls movement.	Remotely piloted model aircraft.
2. Human-Delegated	Human selects specific automated functions.	Automatic take-off or position hold.
3. Human-Supervised	System acts within human-directed tasks; human supervises.	Supervised autonomous survey flight.
4. Fully Autonomous	System turns goals into tasks; emergency intervention remains possible.	Robot vacuum within a specified indoor task.

Question 2

English main text: 646 words.

(1) Involved modules

The proposed delivery system uses all seven parts of the Q1 framework: hardware and sensors, perception and data processing, positioning and SLAM, mission planning, motion planning, modelling and control, and GCS/HMI with communications (Chen, 2022, p. 21). The unmanned ground vehicle (UGV) provides a mobile ground station and recovery deck. I assume bounded vehicle speed and acceleration, a clear, approximately level deck, adequate payload capacity and battery reserve, and a balcony tray with clear overhead access and a registered visual identifier. These conditions define a practical setting for integrating the methods below.

(2) Required features and example methods

Before departure, the GCS supplies the delivery manifest and operating constraints. A behaviour tree organises execution through conditions, actions and fallback branches (Colledanchise & Ögren, 2018). In this proposal, readiness checks cover propulsion, sensors, communication, payload retention and return energy. The tree advances when observations confirm completion, allowing failed searches or infeasible trajectories to trigger bounded retries or return. This connects mission decisions to feedback from the other modules throughout the flight.

Reliable state estimation is essential from the moving launch onward. An EKF-based visual-inertial estimator combines IMU predictions with camera corrections (Mourikis & Roumeliotis, 2007), while reliable GNSS updates and local SLAM mapping are added in this integration (Cadena et al., 2016). Calibrated observations of known deck-marker points provide relative pose through PnP (Lepetit, Moreno-Noguer, & Fua, 2009). Timestamped estimates are expressed in a common frame, allowing the UAV to maintain the deck's horizontal velocity while climbing clear and then transition into outbound flight.

During transit, I combine A* graph search (Hart, Nilsson, & Raphael, 1968) with constrained minimum-snap polynomial planning (Mellinger & Kumar, 2011; Richter, Bry, & Roy, 2016). The resulting timed references must satisfy obstacle clearance, speed and thrust limits. A rigid-body and rotor-thrust model supports cascaded P/PID control: position and velocity loops request thrust

and attitude, while attitude and rate loops request moments. Velocity and acceleration feedforward assist tracking, and allocation produces four feasible rotor commands (PX4, n.d.). New obstacle observations require the planner to revise the trajectory.

Near the apartment building, a planned facade scan maintains visibility and clearance while camera and range measurements locate candidate balconies, people and obstacles. YOLO provides the detection method (Redmon, Divvala, Girshick, & Farhadi, 2016), with training adapted to balcony imagery. The system then checks the building, floor and apartment specified in the manifest against the registered identifier. This distinction matters because recognising a balcony does not establish the intended recipient. Uncertain identity or an obstructed approach prevents delivery.

Once identity and clearance are confirmed, the UAV hovers above the accessible tray. I propose a short winch that lowers the parcel while load or contact sensing checks that the tray supports it. Release is permitted only after this condition is met, and camera evidence confirms deposition before the mechanism retracts. These physical checks establish delivery beyond merely issuing a release command. The controller then updates its mass-dependent hover thrust to account for the removed payload.

The outcome must also reach the truck. MAVLink carries timestamped vehicle-motion estimates, together with battery and system-status telemetry (MAVLink, n.d.). The proposed application sends a delivery record containing the target identity, outcome, time and evidence, with acknowledgement and retries to verify reception. Report reception and physical delivery remain separate completion conditions; a protocol command acknowledgement alone establishes neither. High-rate stabilisation stays onboard, so flight control does not depend on each telemetry exchange.

Recovery closes the mission by updating the moving rendezvous and tracking deck-relative position and velocity (Borowczyk et al., 2017). My design permits descent only with alignment, low horizontal relative speed and reliable tracking. During descent, it continues to match the deck's horizontal motion, reducing vertical relative speed towards zero as it reaches touchdown. Stale updates, marker loss or excessive deck motion require a go-around or safe alternative within the energy reserve. These gates and tolerances require platform-specific validation; returning to the original launch coordinates would miss the moving UGV.

Table 2: Module-to-method mapping for the proposed delivery design. Sources for the named methods appear in the main text; delivery interlocks and integration rules are proposed choices.

Module	**Required feature**	**Example method**
Hardware and sensors	Loaded flight, controlled deposition and physical confirmation.	Winch with load/contact release interlock.
Perception/data processing	Candidates, obstacles, target identity and evidence.	Adapted YOLO plus registered identifier check.
Positioning/SLAM	Vehicle state, map and deck-relative pose.	EKF; calibrated marker PnP near the deck.
Mission/task planning	Sequencing, action conditions and recovery choices.	Behaviour tree with bounded retry/fallback.
Motion planning	Collision-free, dynamically feasible timed references.	A* and constrained minimum-snap trajectory.
Modelling/control	Stable tracking and feasible motor commands.	Cascaded P/PID, feedforward and allocation.
GCS/HMI, communication and results	Mission input, vehicle updates and received delivery records.	MAVLink plus application acknowledgement.

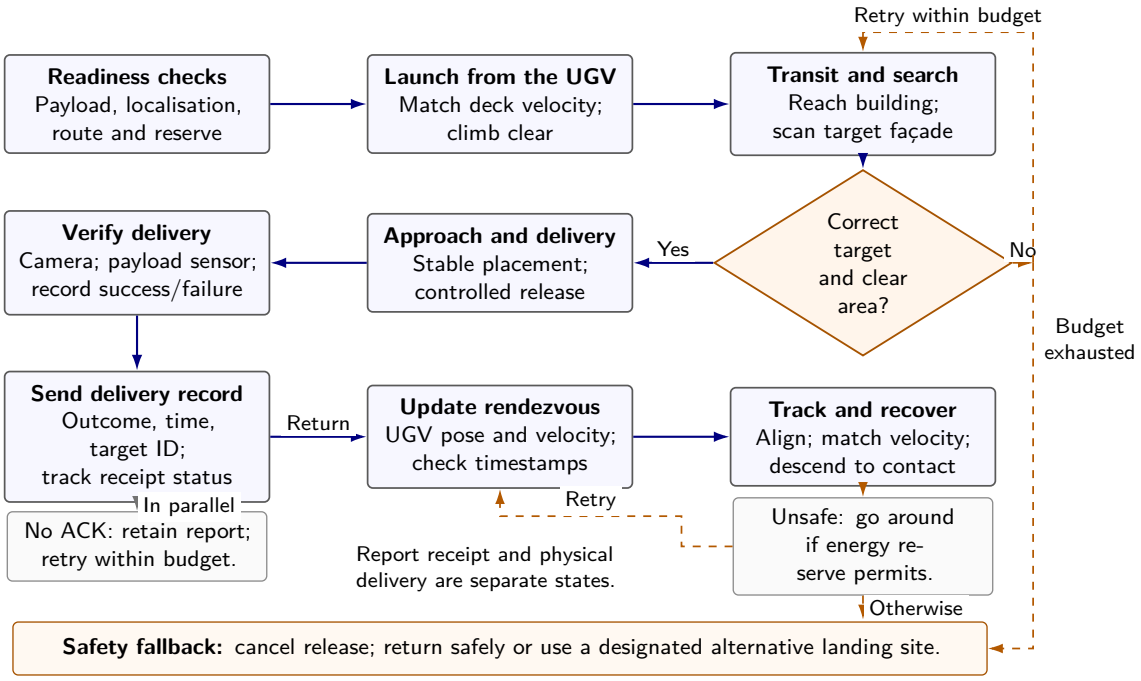


Figure 2: Proposed mission flow with verification and fallback conditions. The task logic uses behaviour-tree principles (Colledanchise & Ögren, 2018); this diagram is a mission flowchart. Physical delivery and reception of its report are separate states.

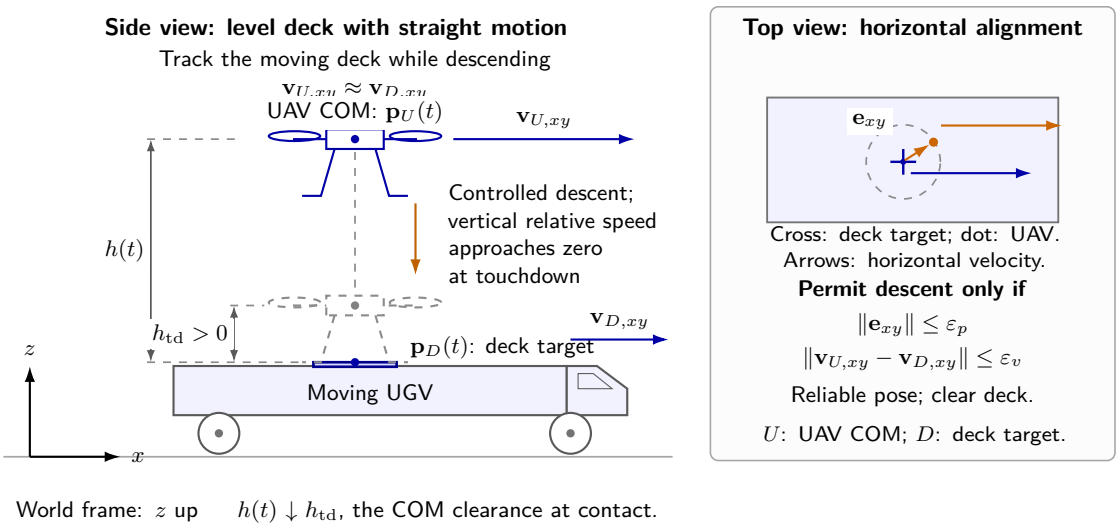


Figure 3: Proposed moving-deck recovery in a common, locally level frame. Mobile-target tracking and camera-pose estimation draw on Borowczyk et al. (2017) and Lepetit et al. (2009). Horizontal matching and a tapered touchdown speed are design choices for this scenario; the touchdown height of the centre of mass remains positive.

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